Compiled Problems – Intermediate Division

1. **00-01 C4 Data Structures**

   Build a binary search tree with the keys RANSOMEVERGLADES (in that order). What is the depth of the tree?

2. **00-01 C4 Data Structures**

   In an initially empty queue, if the following operations are performed (reading down), what letter(s) are left in the queue?

Push (S)	Pop(X)
Push (E)	Pop (X)
Push (C)	Push (E)
Pop (X)	Push (T)
Pop (X)	Pop (X)
Push (R)	

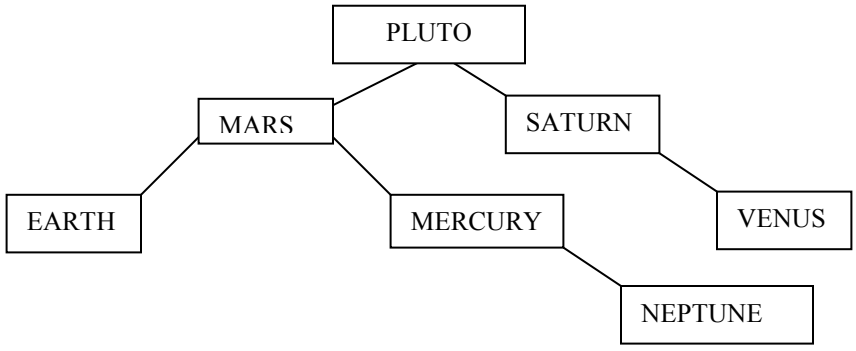
3. **01-02 C4 Data Structures**

   Consider the following operations on an initially empty queue. If a POP is now performed, what value would be popped?

PUSH (B)	PUSH (D)
PUSH (I)	POP (X)
PUSH (N)	PUSH (S)
POP (X)	POP (X)

4. **01-02 C4 Data Structures**

   Insert the key "VULCAN" into the following binary search tree.
   What is the new external path length?



5. **02-03 C4 Data Structures**

Build a binary search tree with the letters **LADYANDTHETRAMP** starting with the letter L and ending with the letter P. What is the depth of the tree?

6. **02-03 C4 Data Structures**

Given an initially empty stack,
what elements are in the stack after the following operations are performed?

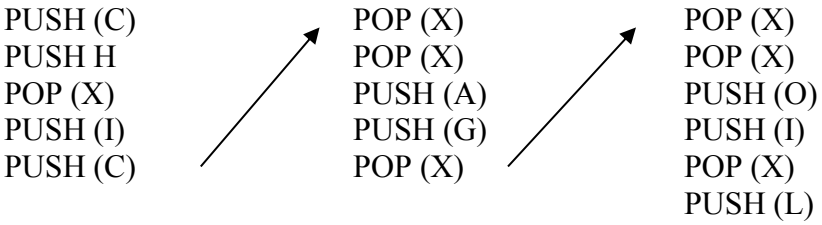
PUSH (S), PUSH (N), PUSH (O), POP (X), PUSH (W), PUSH (W), POP (X),
PUSH (H), PUSH (I), PUSH (T), POP (X), POP (X), PUSH (E), POP (X)

7. **03-04 C4 Data Structures**

Construct a binary search tree for **ELIWHITNEY**. How many nodes have only one child?

8. **03-04 C4 Data Structures**

Consider the following sequence of operations on an initially empty queue.
What items remain in the queue after the operations are executed.



9. **04-05 C4 Data Structures**

What is the depth of the binary search tree for **PHILADELPHIA**?

10. **04-05 C4 Data Structures**

Given an initially empty queue and the following operations, what element would be popped next?

PUSH (L), PUSH (I), PUSH (B), POP (X),
PUSH (E), POP (X), POP (X), PUSH (R)
PUSH (T), POP (X), PUSH (Y)

11. **05-06 C4 Data Structures**

Find the depth of the binary search tree for **OBJECTORIENTED**.

12. **05-06 C4 Data Structures**

What would be the next item removed from the queue constructed by the following commands?

PUSH(I), PUSH(N), PUSH(T), POP(X), PUSH(E), POP(X), PUSH(R), PUSH(M), PUSH(E), POP(X), POP(X), PUSH(D), POP(X), POP(X), PUSH(I), PUSH(A), POP(X), PUSH(T), PUSH(E), POP(X), POP(X).

13. **06-07 C4 Data Structures**

Find the depth of the binary search tree for **THEEYESOFTEXASAREUPONYOU**.

14. **06-07 C4 Data Structures**

What would be the next item removed from the queue constructed by the following commands and noting that REV reverses the order of the elements in the queue?

PUSH(L), PUSH(O), PUSH(N), REV, POP(X), PUSH(E), REV, POP(X), PUSH(S), PUSH(T), REV, POP(X), REV, POP(X), PUSH(A), PUSH(R), REV, POP(X).

15. **07-08 C4 Data Structures**

In the binary search tree for **EDGARALLENPOE**, list all nodes that have 2 children?

16. **07-08 C4 Data Structures**

What would be the next item popped from the queue constructed by the following commands?

PUSH(T), PUSH(H), PUSH(E), POP(X), PUSH(R), PUSH(A), POP(X), POP(X), PUSH(V), PUSH(E), POP(X), PUSH(N), POP(X), POP(X)

17. **08-09 C4 Data Structures**

What is the internal path length of the binary search tree for **HUNTSVILLE**?

18. **08-09 C4 Data Structures**

Define the function REV to reverse the order of the items in the current stack before continuing. What would the next popped element be in the following stack?

PUSH(T), PUSH(R), PUSH(E), REV, POP(X), PUSH(A), PUSH(S), REV, POP(X), POP(X), PUSH(U), PUSH(R), REV, PUSH(E), POP(X), REV

19. **09-10 C4 Data Structures**

What is the depth of the binary search tree for **CONRADWEISERHS**?

20. **09-10 C4 Data Structures**

Define a new operation **CRC(x):** circulate the next x items to be popped to the other end of the structure. Given an initially empty stack and the following sequence of operations, what would be the next POPPED element?

PUSH(A), PUSH(B), PUSH(C), PUSH(D), CRC(2), POP(X), PUSH(E), PUSH(F), CRC(3), POP(X), POP(X)

21. **10-11 C4 Data Structures**

What is the depth of the binary search tree for **TELEPHONEWORKER**?

22. **10-11 C4 Data Structures**

The function SW switches a stack to a queue or a queue to a stack before continuing. What would the next popped element be in the following queue?

PUSH(P), PUSH(L), POP(X), PUSH(A), SW, POP(X), PUSH(N), SW, POP(X), PUSH(N), SW, POP(X), PUSH(E), PUSH(S), SW

23. **11-12 C4 Data Structures**

How many nodes have only 1 child in the binary search tree for **KEYSTONESTATE**?

24. **11-12 C4 Data Structures**

Given an initially empty stack and the following sequence of operations, what would be the next POPPED element?

PUSH(G), PUSH(E), PUSH(T), POP(X), PUSH(T),
POP(X), POP(X), PUSH(Y), PUSH(S), PUSH(B), PUSH(U),
POP(X), PUSH(R), PUSH(G), POP(X), POP(X), POP(X)

25. **12-13 C4 Data Structures**

What is the depth of the binary search tree for **TARHEELSTATE**?

26. **12-13 C4 Data Structures**

What would the next popped element be in the following stack?

PUSH(R), PUSH(A), PUSH(L), PUSH(E), POP(X), POP(X),
PUSH(I), PUSH(G), PUSH(H), POP(X), PUSH(D), PUSH(U),
POP(X), POP(X), PUSH(R), PUSH(H), POP(X), PUSH(M),
POP(X), POP(X), POP(X)

27. **13-14 C4 Data Structures**

How many nodes have only one child in the binary search tree for **DENVERBRONCOS**?

28. **13-14 C4 Data Structures**

What would the next popped element be in the following queue?

PUSH(P), PUSH(I), PUSH(K), POP(X), POP(X), PUSH(E),
PUSH(S), PUSH(P), POP(X), POP(X), PUSH(E), POP(X),
PUSH(A), PUSH(K)

29. **14-15 C4 Data Structures**

Create a binary search tree for the string **WALTDISNEYWORLD**. What is the depth of the tree?

30. **14-15 C4 Data Structures**

Given that the command SWITCH changes a stack to a queue or vice versa and REVERSE switches the order of a given stack or queue, what is the next item to be POPPED from an initially empty queue?

PUSH(S); PUSH(U); PUSH(N); SWITCH; PUSH(S); PUSH(H);
PUSH(I); REVERSE; PUSH(N); PUSH(E); SWITCH; PUSH(S);
PUSH(T); PUSH(A); PUSH(T); PUSH(E); POP(X); POP(X);
SWITCH; POP(X); POP(X); REVERSE; POP(X); POP(X)

31. **15-16 C4 Data Structures**

Find the depth of the binary search tree for **NEWHAMPSHIRE**.

32. **15-16 C4 Data Structures**

What would the next popped element be in the following stack?

PUSH(C), PUSH(O), POP(X), PUSH(N), PUSH(C), POP(X), PUSH(O), PUSH(R), POP(X), POP(X), PUSH(D), PUSH(N), POP(X), PUSH(H), POP(X), POP(X)